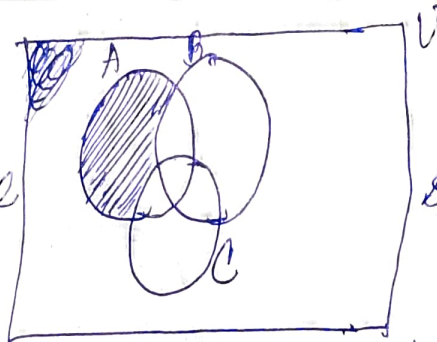


(c) $(A \cap B) \cup (A \cap C)$



sets A and B

Fr 3) What can you say about the
if we know that

of a) $A \cup B = A$

or Solution: $\therefore (A \cup B) \subseteq A$ and $A \subseteq (A \cup B)$. Prob 16(b) already justifies
(a) that $A \subseteq (A \cup B)$.

Fr Let, x be an arbitrary element of $(A \cup B)$, i.e. $x \in (A \cup B)$. By Defn. 3 from
Section 2.1, $x \in A$.

19) By Defn. 1, $\therefore x \in (A \cup B)$, $\therefore x \in A \vee x \in B$. By defn. of OR, at least one
of $x \in A$ or $x \in B$ is True. Let, $x \in B$ be True. By defn. of conditional,
 $(x \in B \rightarrow x \in A)$ is True. By Defn. 3 from Sec-2.1, $\boxed{B \subseteq A} \square$

[So]

b) $A \cap B = A$

De Solution: $\therefore A \cap B \subseteq A$ and $A \subseteq A \cap B$. By Prob 16(a), we know $(A \cap B) \subseteq A$ is
always True.

Let, x be an arbitrary element of A , i.e. $x \in A$. By Defn. 3 from Sec-2.1,
 $x \in A \cap B$. By Defn. 2, $x \in A \wedge x \in B$. By Defn. of AND, $x \in A$ and $x \in B$ both holds.
We already know that $x \in A$ is True.
By defn. of conditional, $x \in A \rightarrow x \in B$ is True. By Defn. 3 from Section 2.1,

$\boxed{A \subseteq B} \square$

c) $A - B = A$

Solution: $\therefore A - B \subseteq A$ and $A \subseteq A - B$. By Prob 16(c), we know that
 $A - B \subseteq A$ always holds.

Let, x be an arbitrary element of A , i.e. $x \in A$.
By Defn. 3 from Sec-2.1, $x \in (A - B)$. By Defn. 4, $x \in A \wedge x \notin B$. By Defn.
of AND, $x \in A$ and $x \notin B$ are both True. We already know that $x \in A$ holds.
Since, $x \notin B$, By Defn. 5, $x \in \bar{B}$. By defn. of conditional, $x \in A \rightarrow x \in \bar{B}$.
By Defn. 3 from Section 2.1, $\boxed{A \subseteq \bar{B}} \square$